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Domain: Artificial Intelligence & Machine Learning

Submitted to : Mr. Mohazin Zahid

Experiment 4

Lab Tasks



Experiment 4.py

Viva Questions

1. Machine learning is different from traditional programming because in traditional programming we write explicit rules and the computer follows those rules to produce outputs, while in machine learning we give the computer examples of data and the correct answers and the computer learns the rules and patterns on its own from the data. In traditional programming, the programmer defines exactly how to solve the problem, whereas in machine learning, the algorithm discovers the solution by finding patterns in the data.
2. The three main types of machine learning are supervised learning, unsupervised learning, and reinforcement learning. Supervised learning uses labeled data where we know the correct answers, and it includes regression for predicting numbers and classification for predicting categories. Unsupervised learning uses unlabeled data to find hidden patterns or groups, like customer segmentation. Reinforcement learning is about learning through trial and error where an agent receives rewards or penalties for its actions, like training a computer to play games.
3. A feature is an input variable used to make a prediction, while a label is the output we want to predict. For example, in a house price prediction problem, features would be things like area of the house, number of bedrooms, and

location, while the label would be the actual price of the house. The model learns the relationship between features and the label during training.

4. We split data into training and test sets because we want to check if our model can work on new, unseen data. If we evaluate the model on the same data it was trained on, it might have just memorized the examples instead of actually learning the pattern, which gives us a false sense of accuracy. The test set acts like new data that the model has never seen before, so we can measure how well it will perform in the real world.
5. The seven steps of the ML workflow are: collect data relevant to the problem, clean and preprocess it by handling missing values and scaling features, split the data into training and test sets, train a model on the training set, evaluate the model on the test set using appropriate metrics, tune hyperparameters to improve performance, and finally deploy the model for real users to use.
6. Classification is used when the output we want to predict is a category or class, like spam or not spam, or what type of flower a plant is, while regression is used when the output is a continuous number, like predicting house prices, temperature, or exam scores. The key difference is that classification predicts discrete categories while regression predicts continuous numbers.

MCQ's

Q1. Predicting a house price is an example of:

(b) Regression

Q2. Labeled data is used in:

(a) Supervised learning

Q3. The output a model predicts is called the:

(b) label

Q4. We evaluate a model on the:

(b) test set

Q5. Grouping customers without labels is:

(c) Clustering

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Intern Signature:

A handwritten signature in black ink, appearing to read 'Ghazi', with a stylized flourish at the end.